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Wireless remote indoor sensor for home automation

Abstract

A heating, ventilation, and air conditioning (HVAC) system includes a network of wireless remote climate sensors to develop a complete heat map of an enclosed space. The remote climate sensor is configured to collect temperature and humidity data on a zone of the enclosed space. The HVAC system uses a network of these sensors to obtain data points across the enclosed space. The resulting heat map is used by the HVAC system to determine where to direct air in the enclosed space. By comparing the temperature and humidity at a specific remote climate sensor with the user's desired temperature and humidity, the HVAC system can decide whether to increase or decrease the air flow through a variable damper that is located near the remote climate sensor. By conducting this analysis throughout the enclosed space and making incremental adjustments to the air flow in hot and cold spots in the enclosed space, the disclosed HVAC system provides even comfort to the user along with reduced energy consumption.

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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 16/661,529, filed Oct. 23, 2019, entitled “Wireless Remote Indoor Sensor for Home Automation”, now U.S. Pat. No. 11,821,645 issued Nov. 21, 2023, which claims priority benefit to provisional application No. 62/749,963 filed Oct. 24, 2018 entitled, “Wireless Remote Indoor Sensor for Home Automation” which are all incorporated herein by reference.

TECHNICAL FIELD

(1) This disclosure relates generally to generally to heating, ventilation, and air conditioning (HVAC) systems and methods of their use. More specifically, this disclosure relates to a wireless remote indoor sensor for HVAC automation.

BACKGROUND

(2) Heating, ventilation, and air conditioning (HVAC) systems are used to regulate environmental conditions within an enclosed space. A thermostat may connect to one or more HVAC units to move, cool, or heat air. Decisions on whether to increase or decrease airflow were traditionally made based on a temperature reading at the thermostat. Because airflow was increased or decreased throughout the system based on a temperature reading in only one isolated area of the enclosed space, hot or cool spots developed in other areas of the enclosed space. Dampers were introduced to restrict air flow to individual rooms to help remedy this shortcoming. However, damper users had to manually adjust the dampers in each room.

(3) Automated dampers were eventually introduced. These automated dampers could include a temperature sensor to help provide a more accurate picture of local temperatures throughout a HVAC system. While an improvement over HVAC systems that make air flow decisions based on a single temperature reading at the thermostat, there are several shortcomings in HVAC systems making heating and cooling decisions based on a network of damper-associated sensors. This approach still gives a vague picture of the temperature map within the enclosed space. For example, a room in a house may only have a single damper. The single sensor cannot illustrate temperature imbalances across the room. Additionally, the location of the sensor at the damper is more accurately describing the temperature of the air exiting the damper rather than the ambient temperature that will be felt by occupants of the room. These factors result in air flow decisions that can exacerbate hot and cool spots in the enclosed space.

SUMMARY OF THE DISCLOSURE

(4) In an embodiment, a heating, ventilation, and air conditioning (HVAC) system includes a network of wireless remote climate sensors to develop a complete heat map of an enclosed space. The remote climate sensor is configured to collect temperature and humidity data on a zone of the enclosed space. The HVAC system uses a network of these sensors to obtain data points across the enclosed space. The resulting heat map is used by the HVAC system to determine where to direct air in the enclosed space. By comparing the temperature and humidity at a specific remote climate sensor with the user's desired temperature and humidity, the HVAC system can decide whether to increase or decrease the air flow through a variable damper that is located near the remote climate sensor. By conducting this analysis throughout the enclosed space and making incremental adjustments to the air flow in hot and cold spots in the enclosed space, the disclosed HVAC system provides even comfort to the user along with reduced energy consumption.

(5) The integration of the wireless remote climate sensors with an HVAC system also permits the creation of personalized microclimates within the enclosed space. In addition to collecting temperature and humidity data, the wireless remote climate sensors can detect whether the enclosed space is occupied by a human. Human detection is made possible by optional cameras, microphones, and gas sensors on the wireless remote climate sensors. As the human moves throughout the enclosed space, the HVAC system is able to track the human's movement using the wireless remote climate sensors. The HVAC system may adjust airflow to different portions of the enclosed space based on the human's location. The result is an efficient use of system resources to keep users at their ideal temperature.

(6) Certain embodiments may include none, some, or all of the above technical advantages. One or more other technical advantages may be readily apparent to one skilled in the art from the figures, descriptions, and claims included herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) For a more complete understanding of the present disclosure, reference is now made to the following description, taken in conjunction with the accompanying drawings, in which:
- (2) FIG. 1 is a diagram of a residential HVAC system incorporating a network of wireless sensors;
- (3) FIG. 2 is a diagram of a wireless HVAC sensor;
- (4) FIG. 3 is a diagram illustrating how the system illustrated in FIG. 1 can be used to maintain an indoor climate;
- (5) FIG. 4 is a flowchart of example methods for maintaining an indoor climate using the system illustrated in FIG. 3;
- (6) FIG. 5 is an example of the system illustrated in FIG. 1 that can create personalized microclimates;
- (7) FIG. 6 is a flowchart of example methods for creating personalized microclimates using the system illustrated in FIG. 5;
- (8) FIG. 7 is a diagram illustrating the integration of digital personal assistants with the system illustrated in FIG. 1;
- (9) FIG. 8 is a diagram illustrating the service bridge functionality of the system illustrated in FIG. 1.

DETAILED DESCRIPTION

(10) Embodiments of the present disclosure and its advantages are best understood by referring to FIGS. 1 through 8 of the drawings, like numerals being used for like and corresponding parts of the various drawings.

(11) As described above, previous HVAC systems lacked the ability to accurately map temperatures across an enclosed space. The present disclosure details a climate sensor designed for installation with existing HVAC systems using a smart thermostat. These sensors form a wireless network that can provide an accurate climate map throughout the enclosed space. The disclosed HVAC system can leverage this network of sensors to make improved allocations of HVAC resources and eliminate hot and cool spots in the enclosed space. A further advantage of the sensor network is that the disclosed HVAC system can track users through different zones of the enclosed space and create a microclimate in the occupied zone while conserving system resources in the unoccupied zones of the enclosed space.

HVAC Sensor Network

(12) FIG. 1 shows an example of a home employing remote sensor network **100** to control the temperature and humidity inside a building. Sensor network **100** is comprised of HVAC system **102**, located on the first floor of the pictured building, and HVAC system **104**, located on the second floor of the pictured building. Systems **102** and **104** are linked to HVAC hardware **106** by air ducts **108**. HVAC hardware **106** conditions air for delivery to a conditioned space. The conditioned space may be, for example, a room, a house, an office building, a warehouse, or the like. In some embodiments, the HVAC hardware **106** is a rooftop unit (RTU) that is positioned on the roof of a building and the conditioned air is delivered to the interior of the building. In other embodiments, portion(s) of the system may be located within the building and portion(s) outside the building.

(13) Generally, HVAC hardware **106** includes furnace **110**, heat exchanger(s) **112**, evaporator(s) **114**, condensing unit(s) **116**, and working-fluid conduit(s) **118**. Furnace **110** heats up heat exchanger(s) **112**. In turn, heat exchanger(s) **112** warm air before it enters air ducts **108** and is delivered to HVAC systems **102** and **104**. Furnace **110** may use any of a number of heat sources. For example, furnace **110** might burn natural gas, propane, oil, or any other combustible compound. Alternatively, furnace **110** might use electric resistance or geo-thermal heat.

(14) An evaporator **114** is generally any heat exchanger configured to provide heat transfer between air flowing through (or across) the evaporator **114** (i.e., air contacting an outer surface of one or more coils of the evaporator **114**) and working fluid passing through the interior of the evaporator **114**. The evaporator **114** may include one or more circuits. Working fluid generally flows from an evaporator **114** to a condensing unit **116** through fluid conduit **118**. A portion of the HVAC hardware **106** is configured to move air across an evaporator **114** and into the air ducts **108** as conditioned airflow.

(15) Working-fluid conduit **118** facilitates the movement of a working fluid (e.g., a refrigerant) through a cooling cycle. The working fluid may be any acceptable working fluid including, but not limited to, fluorocarbons (e.g. chlorofluorocarbons), ammonia, non-halogenated hydrocarbons (e.g. propane), hydrofluorocarbons (e.g. R-410A), or any other suitable type of refrigerant.

(16) A condensing unit **116** is generally comprised of a compressor, a condenser, and a fan. In some embodiments, a condensing unit **116** is an outdoor unit while other components of HVAC hardware **106** may be located indoors. The condensing unit **116** is configured to facilitate movement of the working fluid through the working-fluid conduit **118**. The condenser is generally located downstream of the compressor and is configured to remove heat from the working fluid. The fan is configured to move air across the condenser. For example, the fan may be configured to blow outside air through the condenser to help cool the working fluid flowing therethrough.

(17) The compressed, cooled working fluid flows from the condenser toward an expansion device. The expansion device is coupled to the working-fluid conduit **118** downstream of the condenser and is configured to remove pressure from the working fluid. In this way, the working fluid is delivered to an evaporator **114** and receives heat from airflow to produce a conditioned airflow that is delivered by air ducts **108** to the conditioned space of HVAC systems **102** and **104**. The HVAC hardware **106** may include additional components or may omit one or more components shown in FIG. 1.

(18) Air ducts **108** are distributed throughout the building and use variable dampers **120** as outlets into the building. Variable dampers **120** are valves used to control air flow out of air ducts **108**. Dampers **120** may comprise a single flap that can rotate about the centerline of the flap. In the closed position, the single flap completely obstructs the flow of air out of a variable damper **120**. As the flap is rotated about its centerline, airflow increases through variable damper **120** until the flap reaches the fully open position. The flap can rotate up to 180 degrees from its initial closed position. This allows variable damper **120** to vary the direction in which the airflow is directed as well as the rate of air flow. Alternatively, variable dampers **120** may comprise several rotatable blades instead of a single flap. In the closed position, the blades meet edge-to-edge to completely obstruct the flow of air out of variable damper **120**. Rotation of the blades increases airflow out of variable damper **120**. Rotation of the blades in one direction or the other can be used to change the direction of airflow.

(19) In the present disclosure, any mention of opening or closing a damper does not necessarily mean placing the damper in the fully closed or fully open positions. Opening or closing the damper refers to the act of increasing the degree to which the flap or blades have moved toward the fully open or fully closed positions.

(20) Movement of the flap or blades in a variable damper **120** is performed by a motor. Each variable damper **120** also includes a temperature and humidity sensor. An integrated radio allows variable dampers **120** to wirelessly communicate with a control unit **124**, which is described in detail below. Control unit **124** wirelessly controls the motor in damper **120** to alter the position of the flap or blades. The temperature and humidity data collected at variable damper **120** is wirelessly transmitted to control unit **124**.

(21) In addition to variable dampers **120**, HVAC systems **102** and **104** include wireless climate sensors **122** and control units **124**. Climate sensors **122** collect temperature and humidity data. The climate sensors **122** can be installed throughout an enclosed space to provide accurate climate data

across various portions of the enclosed space. The climate sensors **122** are designed so that they may be installed in pre-existing wall boxes. For example, a climate sensor **122** may be installed in an electrical outlet box in place of a standard electrical outlet. Alternatively, a climate sensor **122** may be installed in a light switch box in place of a standard light switch. The climate sensors **122** can wirelessly communicate with the variable dampers **120** and the control unit **124**. Variable dampers **120**, climate sensors **122**, and control unit **124** may communicate with a variety of wireless protocols. For example, the elements in a single HVAC system may communicate using Bluetooth or Wi-Fi. The structure and function of climate sensor **122** is discussed in more detail below with respect to FIG. 2. HVAC systems **102** and **104** include separate control units **124**. The control unit **124** in HVAC system **102** controls the temperature and airflow through HVAC system **102** while the control unit **124** pictured in HVAC system **104** controls the temperature and airflow through HVAC system **104**. While the example in FIG. 1 shows each HVAC system restricted to a single floor of a building, alternate embodiments might use a single control unit **124** to control variable dampers **120** and to interface with climate sensors **122** that are distributed across multiple levels of a building or other enclosed space.

(22) One example of a control unit **124** is a smart thermostat. Smart thermostats are thermostats with wireless networking capabilities. For example, control unit **124** may connect to the internet using Wi-Fi and with other devices using Bluetooth. Alternative protocols utilizing radio or optical frequencies can also be employed. Control unit **124** additionally includes a memory for storing climate profiles and other user settings. Control unit **124** may also access a cloud database for storing climate profiles and user settings.

(23) User settings stored at control unit **124** include a desired temperature. The user settings may also include timers for changing the temperature from a first desired temperature to a second desired temperature. To factor in the effect of humidity on how a temperature feels to a human, the user settings may include a “feels like” temperature. This “feels like” temperature is represented as ET in the following equation: $ET = T_{sub.0} + w \cdot i_{sub.m} \cdot LR \cdot (P_{sub.a} - RH_{sub.s} \cdot P_{sub.ETs})$. The symbols in the “feels like” equation assume the values listed in the tables below.

(24) TABLE-US-00001 Symbol Description Units $T_{sub.ID}$ Indoor Ambient ° F. (sensor or anticipated value) RH Relative Unitless-% Humidity $T_{sub.OD}$ Outdoor ° F. Ambient X Insulation Unitless Quality Ratio Symbol Description Value Units LR Lewis Ratio 205 ° F./PSI $RH_{sub.s}$ Standard RH 0.45 x Insulation Ratio Poor: 0.08 Unitless (User Average: 0.04 selectable) Good: 0.02 $i_{sub.m}$ Clothing Min: 0.37 Unitless insulation Max: 0.45 permeation efficiency Linearly interpolated based on OD temp, clamped to min at low OD ambient (30° F.), and clamped to max value at high OD amb (90°) $w_{sub.s}$ Standard Skin 0.1 Unitless wettedness $T_{sub.o}$ Operative ($(T_{sub.ID} + T_{sub.r})/2$) ° F. Temperature $P_{sub.a}$ Ambient Vapor RH * $P_{sub.as}$ PSI Pressure $P_{sub.as}$ Ambient $6.11 \cdot 10^{\{(7.5 \cdot T_{sub.ID})/(237.7 + PSI \text{ Saturated } T_{sub.ID})\}}$ * 0.02953 * 0.491154 Vapor Pressure Note that for this calculation, $T_{sub.ID}$ is in ° C. $P_{ET,sub.S}$ Saturated $6.11 \cdot 10^{\{(7.5 \cdot ET)/(237.7 + PSI \text{ Vapor } ET)\}}$ * 0.02953 * 0.491154 Pressure at ET Note that for this calculation, ET is in ° C. $T_{sub.r}$ Radiant $T_{sub.ID} + x \cdot (T_{sub.OD} - T_{sub.ID})$ ° F. Temperature w Skin Unitless Wettedness

(25) Since $P_{sub.ETs}$ (saturated vapor pressure at the ET) is not known until the ET is calculated, it is appropriate to use a first guess of 0.5, then calculate the ET, and re-calculate the $P_{sub.ETs}$. This is done repeatedly until the successive calculations of ET converge to the third decimal place.

(26) In addition to user settings, control unit **124** may store climate profiles **126**. Climate profiles **126** allow each potential user to save climate settings that can be applied when that user is present. Each climate profile **126** includes at least one user identifier **128**. For example, the user identifier **128** might be biometric data, such as the user's face, voice, retina, or fingerprint. The user identifier **128** may include a list of wireless devices associated with the user. Each climate profile includes that user's preferred temperature setting **130**. Climate profiles **126** also contain a preemption value **132** to be used when control unit **124** applies an arbitration logic **134** in the scenario where more

than one user with a climate profile **126** is detected in a zone. The preemption value **132** may be any value equal to or greater than one. It is possible that multiple climate profiles **126** have the same preemption value **132**.

(27) The arbitration logic **134** is a setting stored in control unit **124** that may be changed by a user with administrative privileges. The arbitration logic **134** is a set of rules for control unit **124** to apply when it detects multiple users with climate profiles **126**. For example, the arbitration logic **134** might be a rule that the climate profile **126** retrieved by control unit **124** with the lowest preemption value **132** will always have its settings applied regardless of any other occupants who may have a climate profile **126**. Numerous different arbitration rules may be programmed to suit the needs of the users. Additional explanation of how an arbitration logic **134** operates is provided below with respect to FIGS. 5 and 6.

(28) Control unit **124** is configured to receive user input in several ways. A user can alter settings in control unit **124** by interacting with a touch screen or with physical control buttons on the control unit **124**. A user can also remotely change settings in control unit **124** through a mobile web application. Additional detail about the function of control unit **124** is provided below in the discussion related to FIGS. 3-6.

(29) Wireless Remote Indoor Sensor

(30) FIG. 2 shows an example of the wireless climate sensors **122** depicted in FIG. 1. Climate sensor **122** includes one or more processors **202**, a memory **204**, and a wireless interface **206**. Processor **202** is configured to operate microphone **208**, camera **210**, environmental data collector **212**, speakers **220**, face plate **222**, and button panel **224**. Wireless climate sensors **122** are designed for installation into existing wall boxes, such as a power outlet box or a light switch box. This allows the wireless climate sensor **122** to tie into existing power electrical lines through power junction **226**.

(31) The one or more processors **202** are configured to process data and may be implemented in hardware or software. For example, the processors **202** may be 8-bit, 16-bit, 32-bit, 64-bit or of any other suitable architecture. The processors **202** may include an arithmetic logic unit (ALU) for performing arithmetic and logic operations, processor registers that supply operands to the ALU and store the results of ALU operations, and a control unit that fetches instructions from memory and executes them by directing the coordinated operations of the ALU, registers and other components.

(32) Memory **204** represents any suitable combination of hardware and software configured to store data. The components of memory **204** may comprise volatile memory and/or non-volatile memory. A volatile memory medium may include volatile storage. For example, the volatile storage may include random access memory (RAM), dynamic RAM (DRAM), static RAM (SRAM), and/or extended data out RAM (EDO RAM), among others. In one or more embodiments, a non-volatile memory may include non-volatile storage. For example, the non-volatile storage may include read only memory (ROM), programmable ROM (PROM), erasable PROM (EPROM), electrically erasable PROM (EEPROM), a magnetic storage medium (e.g., a hard drive, a floppy disk, a magnetic tape, etc.), ferroelectric RAM (FRAM), flash memory, a solid state drive (SSD), non-volatile RAM (NVRAM), a one-time programmable (OTP) memory, and/or optical storage (e.g., a compact disc (CD), a digital versatile disc (DVD), a BLU-RAY disc (BD), etc.), among others. The term “memory medium” may mean a “memory device,” a “memory,” a “storage device,” a “tangible computer readable storage medium,” and/or a “computer-readable medium.”

(33) Memory **204** is generally configured to temporarily store data received from microphone **208**, camera **210**, environmental data collector **212**, and speakers **220**. Memory **204** also stores instructions, executable on processor **202**, for operating microphone **208**, camera **210**, environmental data collector **212**, speakers **220**, face plate **222**, and button panel **224**. This includes a voice digitizer and video processing firmware. Memory **204** may also store various programs such as facial recognition software.

- (34) Wireless interface **206** allows climate sensor **122** to wirelessly send and receive data with other devices, including variable dampers **120** and control units **124**. Wireless interface **206** further allows the climate sensor **122** to join a wireless internet network. Wireless interface **206** may use any suitable wireless or optical communication protocol, including Bluetooth, ZigBee, an 802.11 standard, or any other appropriate protocol.
- (35) Environmental data collector **212** is a sensor array that includes a temperature sensor **214** and a humidity sensor **216**. The humidity sensor **216** may be any type of hygrometer including any capacitive, resistive, thermal, gravimetric or any other suitable hygrometer. The temperature sensor **214** may be any sensor operable to measure the temperature of an environment, including, for example an electronic thermometer. Alternate embodiments of environmental data collector **212** also includes a gas detector **218**. The gas detector **218** may be any sensor suitable for detecting the presence and concentration of a gas. For example, the sensor may be a combustible gas sensor, a photoionization detector, an infrared point sensor, an ultrasonic sensor, an electrochemical gas sensor, or a semiconductor sensor. The gas detector **218** can measure the presence of gases such as carbon dioxide, carbon monoxide, and methane.
- (36) In one embodiment face plate **222** is a device operable to display information and receive user input. For example, face plate **222** may be a touch screen with a liquid crystal or OLED display. The touch screen may be any variety of touch screen, including, for example resistive touch, surface capacitive, projected capacitive, surface acoustic wave touch, or infrared touch. In certain embodiments, touch screen installed as face plate **222** is used to operate a light. When wireless climate sensor **122** is installed in a pre-existing light switch box, wireless climate sensor **122** may serve as an electronic switch. The light wiring is removed from the previous light switch and installed at power junction **226**. A physical switch can be depicted on the screen and wireless climate sensor **122** will alter the lighting based on manipulation of the switch depicted on the screen. Alternatively, the touch screen may respond to hand gestures for adjusting power to the light regardless of what is depicted on touch screen.
- (37) Because wireless climate sensor **122** is also designed to fit in wall boxes other than light switch boxes, face plate **222** may be replaced with one of several functional or decorative faceplates. For example, a solid faceplate may be used to minimize the appearance of wireless climate sensor **122**. In an alternate embodiment, face plate **222** may include electrical outlet sockets.
- (38) Additional details about the function of microphone **208**, camera **210**, speakers **220**, and button **224** are included below in the discussions of FIGS. 3-8.
- (39) Management of an HVAC System with a Wireless Remote Indoor Sensor Network
- (40) FIG. 3 is an embodiment of the HVAC system **102** depicted in FIG. 1. Some variable dampers **120** and climate sensors **122** are individually numbered to better illustrate how HVAC system **102** operates. Other aspects of HVAC system **102** were omitted for simplicity. Each HVAC system may be divided into a plurality of zones. Each zone is associated with a variable damper **120**. In FIG. 3, climate sensors **302** and **304** wirelessly communicate with variable damper **306** to form a first climate zone. Climate sensors **308** and **310** wirelessly communicate with variable damper **312** to form a second climate zone. Climate sensors **314**, **316**, and **318** wirelessly communicate with variable damper **320** to form a third climate zone. Variable dampers **306**, **312**, and **320** are in wireless communication with control unit **124**. Alternately, the climate sensors **302**, **304**, **308**, **310**, **314**, **316**, and **318** may communicate directly with control unit **124** and be linked to a zone at the control unit. For example, a user might pair climate sensors **314**, **316**, and **318** with control unit **124** and edit the zone settings at the control unit **124** to associate climate sensors **314**, **316**, and **318** with the third climate zone (variable damper **320**).
- (41) FIG. 4 provides a flowchart for a method **400** of operating an HVAC system based on input from a network of wireless climate sensors as depicted in FIG. 3. At step **402** the climate sensors in HVAC system **102** collect data on the climate of their location, including a temperature and a

humidity measurement. Thus, the pictured climate sensors **302**, **304**, **308**, **310**, **314**, **316**, and **318** each take a temperature and humidity measurement. Variable dampers, such as the depicted variable dampers **306**, **312**, and **320**, likewise collect temperature and humidity measurements. At step **404**, the collected climate data is sent to control unit **124**. This may occur in at least two ways. First, the climate sensors may wirelessly transmit their collected climate data to their associated variable damper. For example, referring to the third zone (variable damper **320**), climate sensors **314**, **316**, and **318** send their climate data to variable damper **320**. Variable damper then wirelessly transmits the data received from climate sensors **314**, **316**, and **318** along with the temperature and humidity measurements collected by variable damper **320** to control unit **124**. Alternatively, climate sensors **314**, **316**, and **318** may wirelessly transmit their climate data directly to control unit **124** while variable damper **320** wirelessly transmits the temperature and humidity measurements that it collects.

(42) At step **406** the control unit **124** determines whether the HVAC system **102** is set to heat or cool. To illustrate, consider a scenario where the HVAC system **102** is set to cool. Now the control unit **124** proceeds to step **408** where it retrieves the user settings and compares it to the climate data received from the climate sensors and variable dampers. In this example the user setting in control unit **124** is an ambient temperature of 70° F. The temperature measurements received from climate sensors from the first climate zone (climate sensor **302**, climate sensor **304**, and variable damper **306**) are 71° F., from the second climate zone (climate sensor **308**, climate sensor **310**, and variable damper **312**) are 68° F., and from the third climate zone (climate sensor **314**, climate sensor **316**, climate sensor **318**, and variable damper **320**) are 73° F. Control unit **124** determines that climate zones one and three exceed the user settings and that climate zone two is below the user settings. Because the temperature measurements from climate zones one and three exceed the user settings, control unit **124** advances to step **410** and increases the airflow to zones one and three. Control unit **124** accomplishes this by wirelessly transmitting instructions to variable dampers **306** and **320** to increase the degree to which the damper is open. Because the temperature in zone three is greater than the temperature in zone one, variable damper **320** is ordered to open to a greater degree than variable damper **306**. Because the temperature measurements from climate zone two fall below the user settings, control unit **124** advances to step **412** and decreases the airflow to zone two. Control unit **124** accomplishes this by wirelessly transmitting instructions to variable damper **312**.

(43) While the previous example addressed how method **400** operates when a temperature imbalance arises between the user settings and the different climate zones, alternate scenarios may arise where the various climate sensors in a climate zone do not return the same temperature measurement. This indicates an uneven heat map across the localized climate zone. To illustrate, assume that the user setting for temperature is 68° F. and the HVAC system **102** is set to heat. This example will focus on climate zone three (climate sensor **314**, climate sensor **316**, climate sensor **318**, and variable damper **320**) for the sake of simplicity. At step **402** climate sensors **314** and **316** measure the temperature as 70° F. Climate sensor **318** measures the temperature as 67° F. Variable damper **320** measures the temperature as 68° F. At step **404**, the collected climate data is sent to control unit **124** as described in the previous example. Control unit **124** determines that HVAC system **102** is set to heat. Proceeding to step **414**, control unit **124** and compares the sensor measurements to the user setting of 68° F. Because climate sensors **314** and **316** measured the temperature as greater than the user setting, control unit **124** proceeds to step **416** to decrease the air flow to the locations of climate sensors **314** and **316**. Because climate sensor **318** measured the temperature as less than the user setting, control unit **124** proceeds to step **418** to increase airflow to the location of climate sensor **318**. This example deviates from the previous example in that control unit **124** must instruct variable damper **320** to do more than just increase the opening of the damper. Control unit **124** must also instruct variable damper **320** on which direction to rotate the damper opening so that the flow of air is increased in the direction of climate sensor **318** while the flow of air is decreased to climate sensors **314** and **316**.

(44) Personalized Microclimates

(45) FIG. 5 is an embodiment of the HVAC system **102** depicted in FIG. 1 that illustrates how wireless devices associated with particular users may be used to create personalized microclimates. A microclimate is a temperature and humidity setting that may be applied as users move between zones. For clarity, each variable damper **120** is renumbered to show the individual zones in HVAC system **102**. Each of variable dampers **502**, **504**, **506**, **508**, **510**, **512**, and **514** has at least one wireless climate sensor **122** paired to that damper to form a zone. Reference to a variable damper should be understood to encompass that zone.

(46) FIG. 6 is a flowchart of a method **600** for creating personalized microclimates in HVAC systems like HVAC system **102** depicted in FIG. 5. Starting at step **602**, the HVAC system **102** detects the presence of an occupant or occupants in a zone. Detection may occur in several ways. In FIG. 5, identification is based on a wireless device, the pictured smart watches **516** and **518**. The climate sensors **122** associated with zone **514** periodically search for Bluetooth capable devices. They detect smart watches **516** and **518**. Proceeding to step **604** the wireless climate sensors and variable damper of zone **514** send a device ID for the smart watches **516** and **518** to control unit **124**. Control unit **124** then determines whether the device ID of either smart watch **516** or **518** matches a device listed as a user identifier in a climate profile stored in control unit **124**.

(47) While the user device in FIG. 5 is a smart watch, this method also works with other wireless devices such as mobile phones, tablets, and computers. The design of wireless climate sensor **122** also permits several other methods of detecting occupants that have a climate profile. For example, camera **210** captures images of humans in the vicinity and facial recognition software loaded to memory **204** and executed on processor **202** may identify the humans. Camera **210** may also function as a retinal scanner. In another embodiment microphone **208** monitors the audio conditions in the vicinity and uses voice recognition software loaded to memory **204** and executed on processor **202** to collect a voice signature that may be compared to a voice signature stored as a user identifier in a climate profile. In yet another embodiment the faceplate **222** is a touchscreen that is configured to scan portions of a hand or finger. Environmental data collector **212** may be used to detect the presence of occupants by measuring the carbon dioxide levels in a zone. When the carbon dioxide level exceeds a baseline value, wireless climate sensor **122** determines that humans or other animals are in the vicinity.

(48) Returning to the example in FIG. 5, control unit **124** determines that that there is a match between the device IDs of smart watches **516** and **518** and at least one climate profile. Control unit **124** proceeds to step **610** where it further determines that smart watches **516** and **518** are listed in a first and a second climate profile, respectively. Control unit **124** then advances to step **612** where it must arbitrate between the first climate profile and the second climate profile to decide which settings to apply to zone **514**. Control unit **124** first checks the arbitration logic settings. In this example the system administrator has established a rule that control unit **124** must find the average of the between the climate settings of the detected climate profiles. This set of averaged values is treated as if it was the dominant profile. In this example, assume that the first climate profile includes a preferred temperature of 71° F. and that the second climate profile includes a preferred temperature of 73° F. Applying the prescribed arbitration logic, control unit **124** determines that the appropriate temperature to apply in zone **514** is 72° F., the average of the preferred temperatures of the first and second climate profiles.

(49) This process is straightforward when only a single profile is detected in a zone. The climate settings in that profile will apply. There may be scenarios where humans are detected in a zone but control unit **124** does not find an associated climate profile. For example, a climate sensor **122** might detect a new Bluetooth capable device, indicating that someone entered the zone, but when the control unit **124** receives the device ID for this new Bluetooth capable device the control unit **124** does not find a match with any of the device IDs listed in the climate profiles. Control unit **124** will retrieve a set of default climate settings to apply in cases like this where an occupant enters a

zone but cannot be identified.

(50) Returning to the example of FIG. 5, control unit **124** proceeds to step **618** and determines whether HVAC system **102** is set to heat or cool. In this example, it determines that HVAC system **102** is set to cool. Proceeding to step **620**, the control unit **124** then determines whether the temperature of zone **514** is greater than the applicable climate setting of 72° F. In this example, variable damper **514** and its associated wireless climate sensors **122** detected a temperature of 74° F. Because the temperature of zone **514** is greater than the applicable climate setting of 72° F., control unit **124** proceeds to step **622** and increases the airflow to zone **514**. Control unit **124** causes an increase in airflow to the zone in question in the same manner as described in FIGS. 3 and 4.

(51) While this example assumed that the climate sensors in zone **514** measured a consistent 74° F. across the zone, there will be cases in which there is a temperature gradient across the zone. The methods discussed in FIG. 4 for correcting temperature imbalances when within a zone are equally applicable in the context of creating personalized microclimates.

(52) An alternate embodiment of the HVAC system **102** pictured in FIG. 5 makes HVAC control decisions based on biometric measures rather than climate profiles. For example, HVAC system **102** can take advantage of the ability of a smart watch **516** to measure a user's heart rate. When a user's heart rate, as detected by smart watch **516**, exceeds at least fifty percent of the user's maximum heart rate, the smart watch **516** may send a signal to control unit **124** that the user of smart watch **516** is likely engaged in moderate to vigorous physical activity. Control unit **124** then determines that it needs to increase airflow to the location of the user of smart watch **516**. In this embodiment, control unit **124** would instruct variable damper **514** to increase airflow. The design of the wireless climate sensors makes it possible to practice this embodiment even if the users do not have wearable devices that can collect biometric measurements and wirelessly transmit them to the control unit **124**. For example, environmental data collector **212** may measure the carbon dioxide content of a room. A spike in carbon dioxide concentration may indicate that an occupant is engaged in vigorous physical activity or that new occupants have entered the zone. When the carbon dioxide measurement data sent to control unit **124** indicates such a spike in carbon dioxide in a zone, control unit **124** will determine that it should increase the airflow to that zone.

(53) Voice Applications for the Climate Sensors

(54) FIG. 7 illustrates how a HVAC system incorporating a network of wireless climate sensors can expand the use of voice control across numerous smart devices and digital personal assistants.

System **700** includes climate sensor **122** control unit **124** which are in signal communication over wireless link **706**. Some reference will be made to the details of climate sensor **122** that are detailed in FIG. 2. Wireless link **706** includes any of the wireless data transmission methods discussed with respect to FIG. 2. Control unit **124** is also in signal communication with smart device **708** using a wireless link **706**. This wireless link **706** may be of the same protocol as the wireless link between climate sensor **122** and control unit **124** or it may be another wireless protocol. Smart device **708** may include one or more electronic device that is configured to receive voice commands, including virtual assistants, smart televisions, and other appliances that respond to voice control.

(55) When someone speaks, climate sensor **122** picks up the voice signal using microphone **208**. Voice digitizer module **702** converts the voice into a digital signal and transfers it to transmission module **704** which packetizes and sends the digital voice signal over wireless link **706**, for example using Wi-Fi, to control unit **124**. Control unit **124** relays the digital voice signal to smart device **708**. Smart device **708** receives the digital voice signal and executes the instructions.

(56) The previous example of system **700** requires depends on smart device **708** being able to receive digital commands in place of audible commands. An alternate embodiment of system **700** avoids potential incompatibility issues by leveraging the climate sensor network as an intercom system. The initial steps of voice detection by a climate sensor **122** and processing of the signal are the same. This alternate embodiment deviates from the previous example following the receipt of

the digital voice signal at control unit **124**. Instead of sending the signal directly to the smart device **708**, control unit **124** analyzes the voice signal to determine which climate sensor is nearest to the destination smart device **708**. Control unit **124** then transmits the digital voice signal to that climate sensor **122** using a wireless link **706** such as Wi-Fi. The destination climate sensor **122** receives the digital voice signal, converts the signal to analog using a DAC module, and emits the analog signal through speaker **220**. The audio signal emitted from speaker **220** reaches the smart device **708** as if the user was speaking near the smart device **708**.

(57) Bridge for Integrating HVAC Services with Home Automation Platforms

(58) FIG. **8** illustrates how a HVAC system incorporating a network of wireless climate sensors can serve as bridge for integrating disparate home automation systems. Current systems require that any accessories trying to communicate with a home automation system send the requests in the native language of the home automation system. This makes it difficult to integrate devices designed for different platforms.

(59) System **800** is centered around wireless bridge **802**. Wireless bridge **802** is the network formed by a plurality of wireless climate sensors **122** and a control unit **124**. Details on the structure and function of climate sensors **122** and control unit **124** may be found in the discussion of FIGS. **1-2**. In system **800** the control unit **124** is further configured with a translation module, stored in memory **204** and executable on processor **202**, capable of converting between proprietary home automation protocols. Wireless bridge **802** is in signal communication with cloud database **804** and home automation system **806** through wireless link **808**. A wireless link **808** also allows control unit **124** to communicate with a first virtual assistant device **810**, a second virtual assistant device **811**, and second home automation system **814**. Wireless link **800** is any suitable wireless protocol, including Wi-Fi or Bluetooth.

(60) Cloud database **804** is a remote operating system for an HVAC system. Home automation systems **806** and **814** comprise at least one control device paired with at least one home appliance. For example, home automation system **806** may comprise a mobile control application on a smartphone that controls the operation of lights and ceiling fans in a house. Home automation system **814** comprises a mobile control application on a smartphone that controls televisions and audio appliances in the house. The controllers for home automation systems **806** and **814** are not interoperable.

(61) Virtual assistant devices **810** and **812** are likewise incapable of operating together. Each virtual assistant device **810** and **812** is a software agent embedded in the operating system of a smart device that can perform tasks based on audible commands or questions. Virtual assistant device **810** can control other smart devices using the same operating system as virtual assistant device **810**, and virtual assistant device **812** can control other smart devices using the same operating system as virtual assistant device **812**.

(62) Users can harmonize these disparate home automation systems and virtual assistant devices using wireless bridge **802**. For example, a user provides a command to a virtual assistant device **810**. Virtual assistant device **810** transmits instructions for carrying out the command to control unit **124** of wireless bridge **802**. Control unit **802** determines which that the appliance to which the command applies is controlled by home automation system **814**. Control unit **124** then translates the command from the protocol used by virtual assistant **810** to that of home automation system **814**. Control unit then sends the translated command to home automation system **814**. The command may be sent directly to home automation system **814**, or in the case where home automation system **814** can only send and receive wireless signals from short range, control unit **814** may identify which of the climate sensors **122** is nearest the home automation system **814** and use that climate sensor **122** to relay the command to home automation system **814**.

(63) Modifications, additions, or omissions may be made to the methods described herein without departing from the scope of the disclosure. The methods may include more, fewer, or other steps. Additionally, steps may be performed in any suitable order.

(64) Although this disclosure has been described in terms of certain embodiments, alterations and permutations of the embodiments will be apparent to those skilled in the art. Accordingly, the above description of the embodiments does not constrain this disclosure. Other changes, substitutions, and alterations are possible without departing from the spirit and scope of this disclosure.

(65) While several embodiments have been provided in the present disclosure, it should be understood that the disclosed systems and methods might be embodied in many other specific forms without departing from the spirit or scope of the present disclosure. The present examples are to be considered as illustrative and not restrictive, and the intention is not to be limited to the details given herein. For example, the various elements or components may be combined or integrated in another system or certain features may be omitted, or not implemented.

(66) In addition, techniques, systems, subsystems, and methods described and illustrated in the various embodiments as discrete or separate may be combined or integrated with other systems, modules, techniques, or methods without departing from the scope of the present disclosure. Other items shown or discussed as coupled or directly coupled or communicating with each other may be indirectly coupled or communicating through some interface, device, or intermediate component whether electrically, mechanically, or otherwise. Other examples of changes, substitutions, and alterations are ascertainable by one skilled in the art and could be made without departing from the spirit and scope disclosed herein.

(67) To aid the Patent Office, and any readers of any patent issued on this application in interpreting the claims appended hereto, applicants note that they do not intend any of the appended claims to invoke 35 U.S.C. § 122(f) as it exists on the date of filing hereof unless the words “means for” or “step for” are explicitly used in the particular claim.

Claims

1. A system for managing indoor climates, comprising: a first zone in a room comprising: a first variable damper having a temperature sensor, wherein the first variable damper controls a flow of air through a first air duct outlet, and wherein the temperature sensor of the first variable damper measures temperature of the flow of air through the first air duct outlet to the first zone in the room; a first remote sensor that measures temperature in the first zone of the room; and a second remote sensor that measures temperature in the first zone of the room; a second zone in the room comprising: a second variable damper having a temperature sensor, wherein the second variable damper controls the flow of air through a second air duct outlet, and wherein the temperature sensor of the second variable damper measures temperature of a flow of air through the second air duct outlet to the second zone, a third remote sensor that measures temperature in the second zone of the room, and a fourth remote sensor that measures temperature in the second zone of the room; a control unit wirelessly coupled to the remote sensors and the variable dampers, the control unit comprising: a memory configured to store current climate settings, wherein the current climate settings include a desired temperature for the first zone for at least a first user and a second user, a desired temperature for the second zone for at least the first user and the second user, and a HVAC mode, wherein the control unit is operable to determine an average temperature for each of the first zone and the second zone based on the current climate settings of at least the first user and the second user; a screen configured to display the current climate settings and further configured to receive inputs from a user for changing the current climate settings; and a processor configured to: receive temperature measurements from the first remote sensor, the second remote sensor, and the temperature sensor of the first variable damper of the first zone and the third remote sensor, the fourth remote sensor, and the temperature sensor of the second variable damper of the second zone; retrieve, from the memory, the current climate settings; determine the current climate settings to apply to the first zone and the second zone; compare the temperature measurements from the first

remote sensor, the second remote sensor, and the temperature sensor of the first variable damper from the first zone with the average temperature of the first zone; compare the temperature measurements from the third remote sensor, the fourth remote sensor, and the temperature sensor of the second variable damper from the second zone with the average temperature of the second zone; instruct the first variable damper to alter the flow of air into the first zone if the temperature measurements from the first zone differ from the average temperature of the first zone; and instruct the second variable damper to alter the flow of air into the second zone if the temperature measurements from the second zone differ from the average temperature of the second zone.

2. The system of claim 1, wherein the HVAC mode is heating.

3. The system of claim 2, wherein the temperature measurements of the first zone exceed the average temperature of the first zone; and the temperature measurements of the second zone fall below the average temperature of the second zone.

4. The system of claim 3, wherein the processor instructs the first variable damper to alter the flow of air into the first zone by closing and the processor instructs the second variable damper to alter the flow of air into the second zone by opening.

5. The system of claim 1, wherein the HVAC mode is cooling.

6. The system of claim 5, wherein the temperature measurements of the first zone exceed the average temperature of the first zone; and the temperature measurements of the second zone fall below the average temperature of the second zone.

7. The system of claim 6, wherein the processor instructs the first variable damper to alter the flow of air into the first zone by opening and the processor instructs the second variable damper to alter the flow of air into the second zone by closing.

8. A method of managing indoor climates, the method comprising: measuring a temperature in a first zone in a room with a first remote sensor, a second remote sensor and a temperature sensor positioned in a first variable damper, wherein the first variable damper controls a flow of air through a first air duct outlet, wherein the temperature sensor of the first variable damper measures temperature of the flow of air through the first air duct outlet to the first zone; measuring a temperature in a second zone with a third remote sensor, a fourth remote sensor and a temperature sensor positioned in a second variable damper, wherein the second variable damper controls a flow of air through a second air duct outlet, wherein the temperature sensor of the second variable damper measures temperature of the flow of air through the second air duct outlet to the second zone; retrieving desired climate settings, wherein the desired climate settings include a desired temperature for at least a first user and a second user; determining the desired climate settings to apply to the first zone and the second zone; determining an average temperature for each of the first zone and the second zone based on the desired climate settings of at least the first user and the second user; comparing the temperature measurements from the first zone with the average temperature of the first zone; comparing the temperature measurements from the second zone with the average temperature of the second zone; instructing the first variable damper associated with the first zone to alter the flow of air into the first zone if the temperature measurements in the first zone differ from the average temperature of the first zone; and instructing the second variable damper associated with the second zone to alter the flow of air into the second zone if the temperature measurements in the second zone differ from the average temperature of the second zone.

9. The method of claim 8, wherein the desired climate settings further include an HVAC mode, and wherein the HVAC mode is set to heating.

10. The method of claim 9, wherein comparing the temperature measurements from the first zone with the average temperature of the first zone reveals that the temperature-measurements of the first zone exceed the average temperature of the first zone; and wherein comparing the temperature measurements from the second zone with the average temperature of the second zone reveals that the temperature measurements of the second zone fall below the average temperature of the second

zone.

11. The method of claim 10, wherein instructing the first variable damper associated with the first zone to alter the flow of air into the first zone if the temperature measurements in the first zone differ from the average temperature of the first zone comprises instructing the first variable damper to close; and wherein instructing the second variable damper associated with the second zone to alter the flow of air into the second zone if the temperature measurements in the second zone differ from the average temperature of the second zone comprises instructing the first variable damper to open.

12. The method of claim 8, wherein the desired climate settings further include an HVAC mode, and wherein the HVAC mode is set to cooling.

13. The method of claim 12, wherein comparing the temperatures from the first zone with the average temperature of the first zone reveals that the temperature measurements of the first zone exceed the average temperature of the first zone; and wherein comparing the temperature from the second zone with the average temperature of the second zone reveals that the temperature measurements of the second zone fall below the average temperature of the second zone.

14. The method of claim 13, wherein instructing the first variable damper associated with the first zone to alter the flow of air into the first zone if the temperature measurements in the first zone differ from the average temperature of the first zone comprises instructing the first variable damper to open; and wherein instructing the second variable damper associated with the second zone to alter the flow of air into the second zone if the temperature measurements in the second zone differ from the average temperature of the second zone comprises instructing the first variable damper to close.

15. A controller for managing an indoor climate system, the controller comprising: a memory configured to store: current climate settings, wherein the current climate settings include a desired temperature for a first zone for at least a first user and a second user, a desired temperature for a second zone for at least the first user and the second user, and a HVAC mode, wherein the controller is operable to determine an average temperature for each of the first zone and the second zone based on the current climate settings of at least the first user and the second user; and a screen configured to display the current climate settings, and further configured to receive inputs from a user for changing the current climate settings; and a processor configured to: receive from a first remote sensor, located in a first zone in a room, temperature measurements in the first zone; receive from a second remote sensor, located in the first zone in the room, temperature measurements in the first zone; receive from a temperature sensor positioned in a first variable damper, temperature measurements for a flow of air through a first air duct outlet to the first zone in the room, wherein the first variable damper controls the flow of air through the first air duct outlet; receive from a third remote sensor, located in a second zone in the room, temperature measurements in the second zone; receive from a fourth remote sensor, located in the second zone in the room, temperature measurements in the second zone; receive from a temperature sensor positioned in a second variable damper temperature measurements for a flow of air through a second air duct outlet to the second zone in the room, wherein the second variable damper controls the flow of air through the second air duct outlet; retrieve, from the memory, the current climate settings; determine the current climate settings to apply to the first zone and the second zone; compare each of the temperature measurements received from the first remote sensor, the second remote sensor, and the temperature sensor positioned in the first variable damper from the first zone with the average temperature of the first zone; compare each of the temperature measurements received from the third remote sensor, the fourth remote sensor, and the temperature sensor positioned in the second variable damper from the second zone with the average temperature of the second zone; instruct the first variable damper to alter the flow of air into the first zone if the temperature measurements from one of the first remote sensor, the second remote sensor, or the temperature sensor of the first variable damper differs from the average of the first zone; and instruct the second variable damper

to alter the flow of air into the second zone if the temperature measurements from one of the third remote sensor, the fourth remote sensor, or the temperature sensor of the second variable damper differs from the average temperature of the second zone.

16. The controller of claim 15, wherein the HVAC mode is heating; the temperature measurements received from the first remote sensor and the second remote sensor exceed the average temperature of the first zone; and the temperature measurements received from the third remote sensor and the fourth remote sensor exceed average temperature of the second zone.

17. The controller of claim 16, wherein the processor instructs the first variable damper to alter the flow of air into the first zone by closing and the processor instructs the second variable damper to alter the flow of air into the second zone by opening.

18. The controller of claim 15, wherein the HVAC mode is cooling; the temperature measurements received from the first remote sensor and the second remote sensor exceed the average temperature of the first zone; and the temperature measurements received from the third remote sensor and the fourth remote sensor exceed the average temperature of the second zone.

19. The controller of claim 18, wherein the processor instructs the first variable damper to alter the flow of air into the first zone by opening and the processor instructs the second variable damper to alter the flow of air into the second zone by closing.

20. The controller of claim 15, wherein the desired temperature is a feels like temperature.
